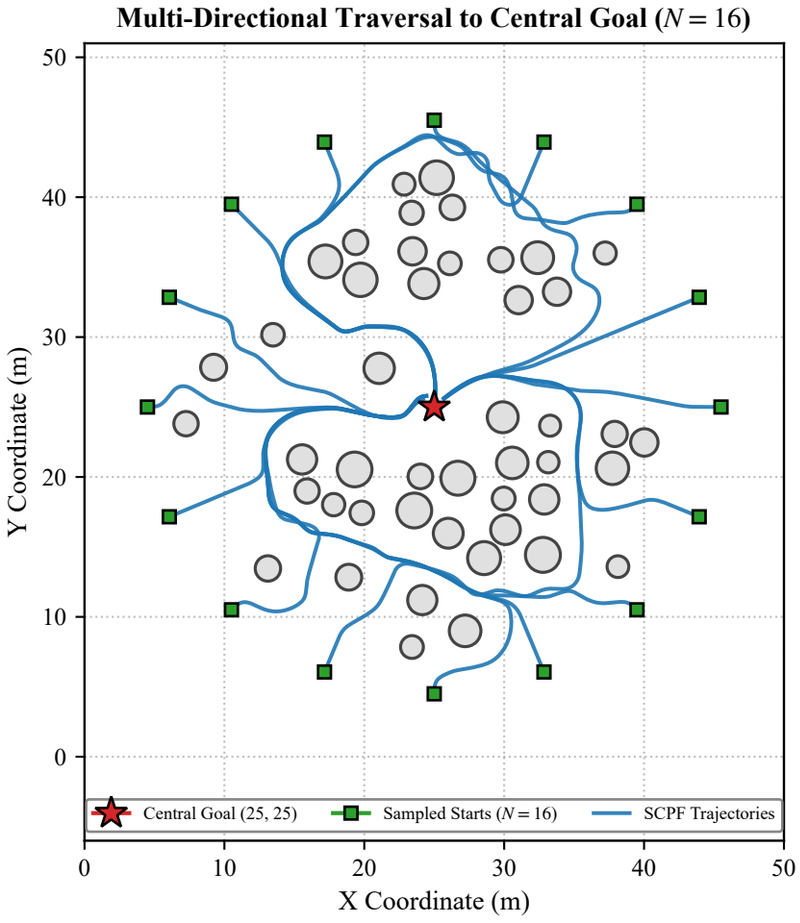
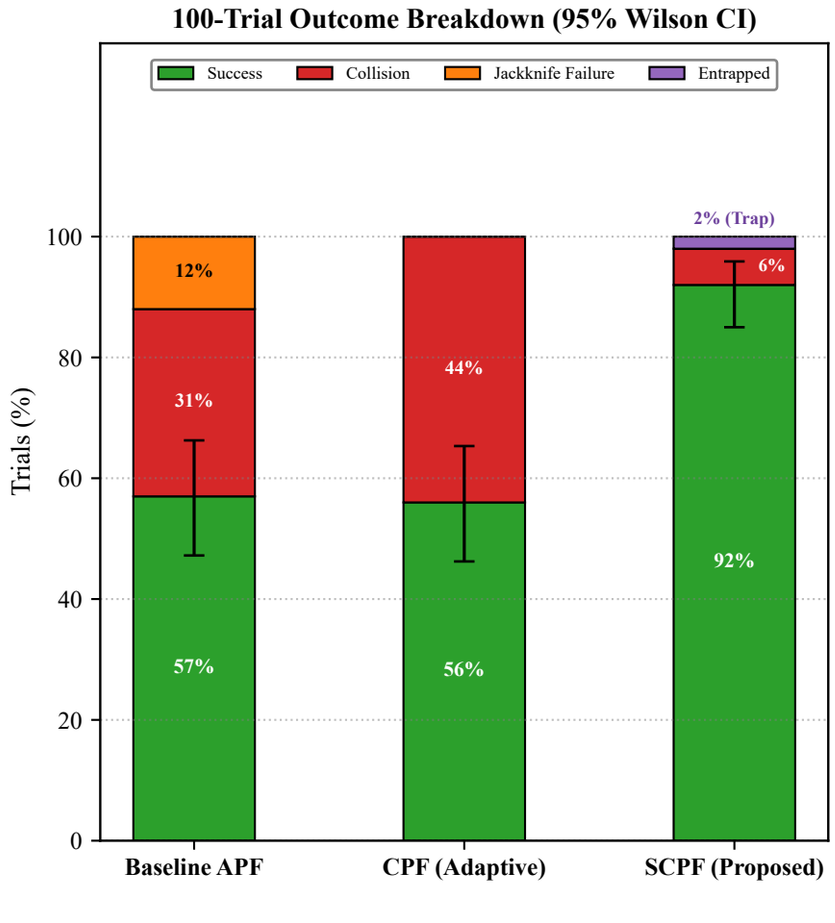
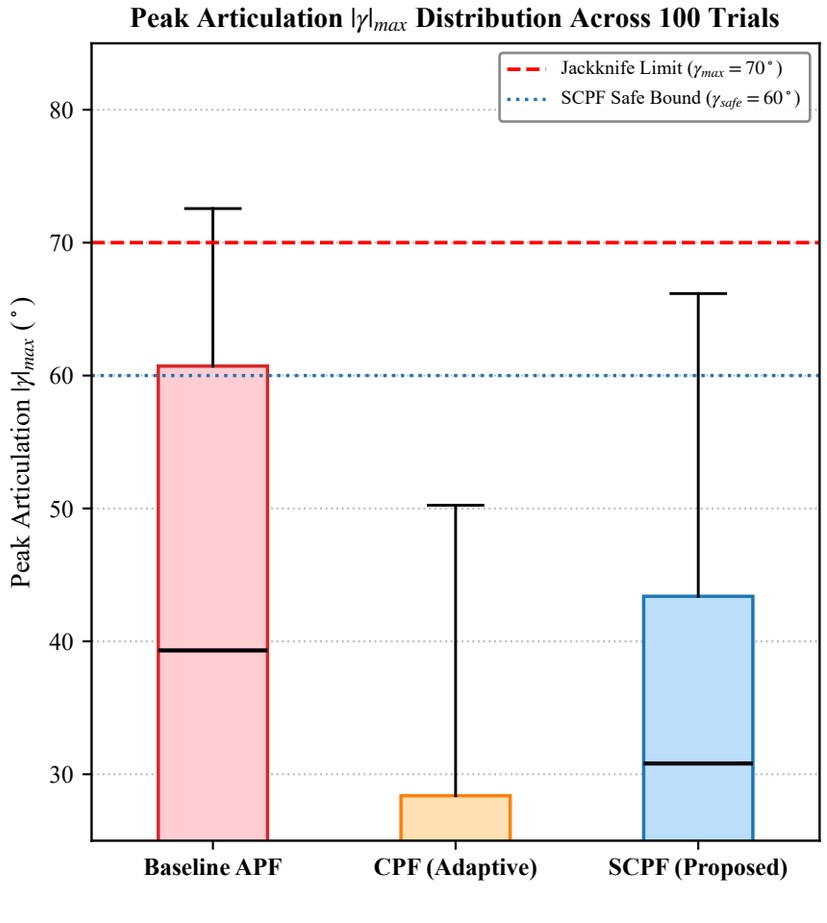

Multi-Directional Traversal to Central Goal ($N = 16$)
Y Coordinate (m)
X Coordinate (m)
0
10
20
30
40
50
Central Goal (25, 25)
Sampled Starts ($N = 16$)
SCPF Trajectories


100-Trial Outcome Breakdown (95% Wilson CI)
Success
Collision
Jackknife Failure
Entrapped
Trials (%)
12%
31%
57%
44%
56%
2% (Trap)
6%
92%
Baseline APF
CPF (Adaptive)
SCPF (Proposed)


Peak Articulation $|\gamma|_{max}$ Distribution Across 100 Trials
Jackknife Limit ($\gamma_{max} = 70^\circ$)
SCPF Safe Bound ($\gamma_{safe} = 60^\circ$)
Peak Articulation $|\gamma|_{max}$ ($^\circ$)
Baseline APF
CPF (Adaptive)
SCPF (Proposed)